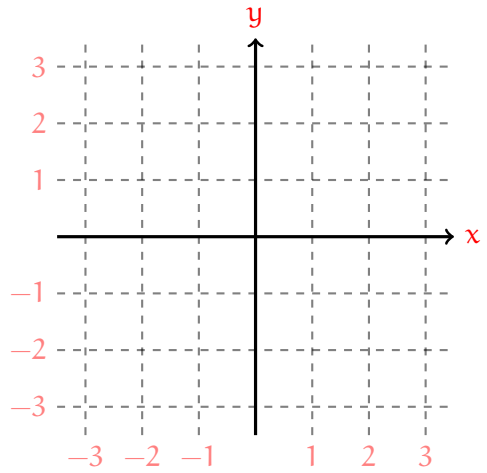
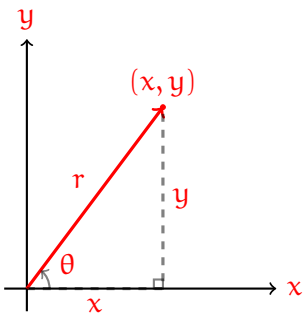
Example 1. Sketch the region determined by $2 \leqslant r \leqslant 3$ and $-\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}$.



Solution. The requirement $2 \leqslant r \leqslant 3$ considers only points

The requirement $-\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}$ considers only points

A. **Polar Conversion.** We convert between standard and polar coordinates.



Key Relations Between Polar and Standard Coordinates.

$x =$

$y =$

$r^2 =$

thus when r is nonnegative:

$y \cos\theta =$

thus when x is nonnegative:

thus when y is nonnegative:

Example 2. Graph $r = 2\cos\theta$ by first converting to standard coordinates.

Solution. To immediately introduce two key relations, we multiply by r.

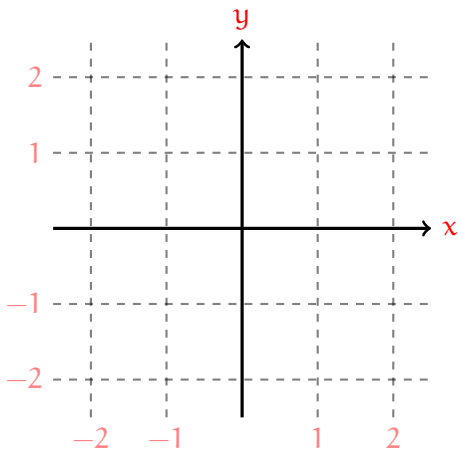
$r = 2\cos\theta$

Next we complete the square with respect to x.

Recall that completing the square utilizes the relation

$x^2 + bx +$

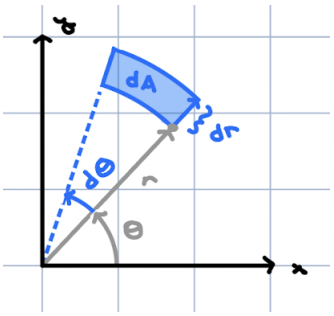
Back to our problem.



B. **Polar Integration.** Now we will discuss polar integration.

Infinitesimal area dA can be obtained from point with polar coordinates r, θ by

- increasing θ by:
- increasing r by:

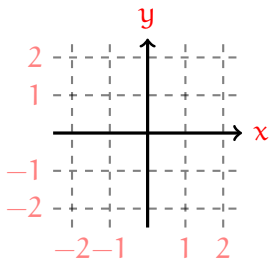


This dA may be approximated by a rectangle with

- base equal to the arclength:
- height equal to:

In polar coordinates:
$dA =$

Example 3. Let D be bounded by $x^2+y^2=1$ and $x^2+y^2=4$ in the 1st quadrant.



Find: $\iint_D e^{x^2+y^2}\, dA$.

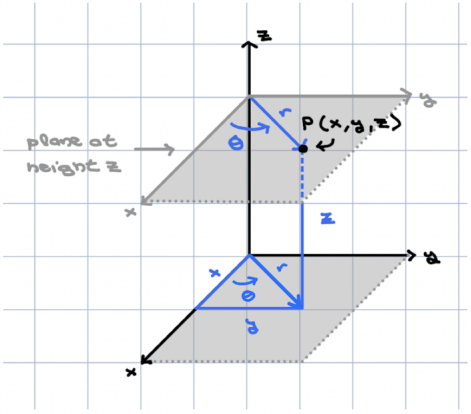
Solution. First we describe the region D using polar coordinates.

Next we use $dA = r\, dr d\theta$ and $r^2 = x^2+y^2$ to convert our integral to polar.

$\iint_D e^{x^2+y^2}\, dA =$

C. **Cylindrical Coordinates.** A point (x, y, z) has **cylindrical coordinates** r, θ, z.

- (r, θ) gives:
- z measures:



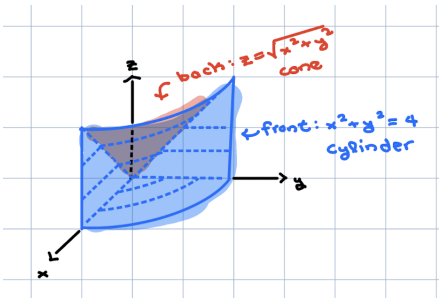
Another convenient viewpoint, when r is nonnegative, is

- r measures:
- sketch the **flat radial vector** from the z–axis to the point, then θ measures

So, for example, the cylindrical equation $r = 5$ defines:

We can also view in Desmos.

Example 4. Let E be the region in the first octant under the cone $z = \sqrt{x^2 + y^2}$ and inside the cylinder $x^2 + y^2 = 4$. Give inequalities in cylindrical coordinates that describe the region.



Solution. For each fixed r and θ satisfying:

The range of z is: